Active Clamping of High-Voltage IGBTs: Theory, Design for a 6.5 kV / 1000 A Module and a Simscape TVS Model

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Abstract

Active clamping is a gate-side overvoltage protection technique in which the collector–emitter voltage of an IGBT is fed back to the gate through a stack of transient voltage suppressor (TVS) diodes. When the collector voltage exceeds the stack breakdown level, the device is forced back into linear conduction and absorbs the inductive commutation energy while its voltage is limited to the clamp level. This note reviews the operating principle and its variants, develops a complete design procedure for the 6.5 kV / 1000 A HiPak module 5SNA 1000G650300, including tolerance, temperature and energy budgets and the coordination with an external MOV clamp, and documents `tvs_clamp`, a Simscape component modelling the TVS stack for time-domain verification together with the `igbt_dyn` switching model.

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1 Introduction

At turn-off, the current commutating out of an IGBT leaves the energy $\frac{1}{2}L_\sigma I_0^2$ stored in the commutation loop inductance L_σ with no path other than the device itself. The resulting overshoot

$$\hat{V}_{CE} \approx V_{dc} + L_\sigma \left| \frac{\mathrm{d}i_C}{\mathrm{d}t} \right| \tag{1}$$

must remain inside the RBSOA. For 6.5 kV-class devices the margin between the nominal DC link (typically 3.6 kV) and the blocking rating is large in absolute terms, but so are the currents, the loop inductances of large mechanical assemblies, and the energies involved. Passive means—decoupling and RCD snubbers [7, 8] and metal-oxide varistors [9]—limit the overshoot by absorbing or diverting the loop energy. Active clamping is complementary: it uses the IGBT itself as the absorbing element, exploiting the fact that a modern high-voltage IGBT can dissipate several joules per event inside its own SOA, and requires only a small signal-level feedback network.

Active clamping is standard practice in high-voltage gate drivers (traction, medium-voltage drives, solid-state switches), both as the primary overshoot limiter and as a last-line protection for abnormal turn-off events such as short-circuit interruption, where I_0 and $\mathrm{d}i/\mathrm{d}t$ far exceed their nominal values [1, 3].

2 Active Clamping: Principle and Variants

2.1 Operating principle

Figure 1 shows the basic arrangement on an inductively loaded low-side switch. A stack of N series-connected TVS diodes is placed between the collector (preferably the auxiliary collector sense terminal) and the gate. As long as v_{CE} is below the stack breakdown voltage $V_{BR,s} = NV_{BR}$, the stack only leaks microamps and the gate loop is unaffected. When, during turn-off, v_{CE} reaches $V_{BR,s}$, the stack avalanches and injects a current i_{TVS} into the gate node against the driver, which is holding the gate low through $R_{g,\mathrm{off}}$. The gate voltage rises to the Miller plateau corresponding to the instantaneous load current, the IGBT re-enters linear conduction, and the collector voltage is held at

$$v_{CE,\mathrm{cl}} \;=\; V_{BR,s} + N r_z \, i_{TVS} + v_{GE,\mathrm{pl}}, \tag{2}$$

where r_z is the dynamic resistance of one TVS device and $v_{GE,\mathrm{pl}}$ the plateau voltage. The clamp is therefore a negative-feedback loop: any tendency of v_{CE} to rise increases i_{TVS}, raises v_{GE}, increases the device conduction and pulls v_{CE} back down.

2.2 Clamped turn-off waveforms and energy

During the clamped interval the voltage across the loop inductance is fixed, so the current decays linearly (Figure 2):

$$\frac{\mathrm{d}i_C}{\mathrm{d}t} = -\,\frac{v_{CE,\mathrm{cl}} - V_{dc}}{L_\sigma}, \qquad \Delta t_{\mathrm{cl}} = \frac{L_\sigma I_0}{v_{CE,\mathrm{cl}} - V_{dc}}. \tag{3}$$

The energy dissipated in the IGBT during the clamp interval is the loop energy plus the contribution drawn from the source while the current decays:

$$E_{\mathrm{IGBT}} \approx \frac{1}{2} v_{CE,\mathrm{cl}} \, I_0 \, \Delta t_{\mathrm{cl}} = \frac{1}{2} L_\sigma I_0^2 \, \frac{v_{CE,\mathrm{cl}}}{v_{CE,\mathrm{cl}} - V_{dc}}. \tag{4}$$

Note the amplification factor $v_{cl}/(v_{cl} - V_{dc})$: a clamp level close to the DC link stretches the decay and multiplies the energy. The TVS stack itself only carries the gate-loop current, so its

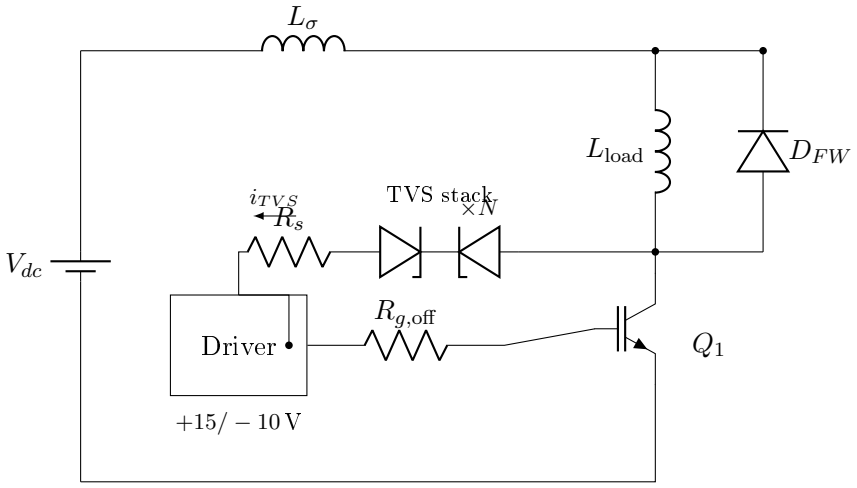


Figure 1: Basic active clamping of a low-side IGBT. The TVS stack (drawn as a bidirectional pair for one of the N devices) feeds the collector overvoltage back into the gate node; R_s is a small series/damping resistor. The stack should sense the auxiliary collector terminal to exclude the main-terminal inductance from the feedback path.

energy is smaller by roughly the ratio i_{TVS}/I_0 — three to four orders of magnitude:

$$E_{TVS} \approx v_{CE,\text{cl}}\, i_{TVS}\, \Delta t_{\text{cl}}. \tag{5}$$

This division of labour — signal-level stress in the clamp network, power-level stress in the IGBT — is the defining feature of active clamping compared with MOV or RCD clamps, where the clamp element absorbs the full energy.

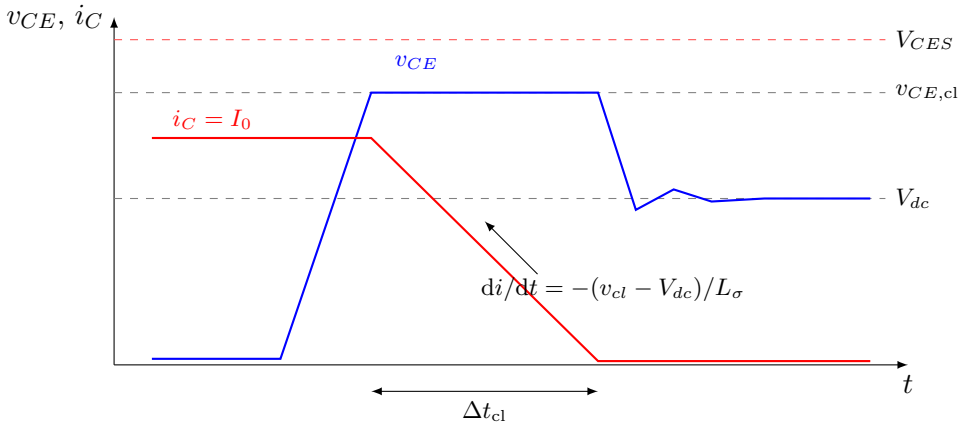


Figure 2: Idealised clamped turn-off. The voltage rises, is caught at $v_{CE,\text{cl}}$ below the rating, and the current ramps down linearly during Δt_{cl} while the IGBT dissipates E_{IGBT}.

2.3 Basic versus advanced active clamping

In the *basic* scheme of Figure 1 the entire feedback current flows through the TVS stack. Its quasi-stationary value is set by the gate network,

$$i_{TVS} \approx \frac{v_{GE,\text{pl}} - V_{EE}}{R_{g,\text{off}}} + C_{ies}\frac{\text{d}v_{GE}}{\text{d}t} + C_{res}\frac{\text{d}v_{CE}}{\text{d}t}, \tag{6}$$

i.e. the current needed to hold the gate at the plateau against the driver pull-down, plus the input-capacitance charging spike at clamp engagement and the Miller displacement current. With the

ohmic turn-off resistors used at the 6.5 kV level this amounts to one to a few amperes, and the term $Nr_z i_{TVS}$ in (2) raises the clamp level substantially above the breakdown threshold: the *clamping factor* $v_{cl}/V_{BR,s}$ of small axial TVS devices operated near their rated pulse current is 1.3–1.4, which, as Section 3 shows, consumes most of the margin between the DC link and V_{CES}.

In *advanced active clamping* (AAC), as implemented in commercial high-voltage driver cores [4], the TVS stack only carries a small sense current (of the order of 0.1–0.5 A) into a dedicated driver input; an auxiliary stage inside the driver amplifies it, partially or fully disabling the pull-down path (and possibly actively sourcing gate current). The stack then operates just above its knee, the clamping factor drops to ≈ 1.05, and both the TVS pulse stress and the spread of the clamp level shrink accordingly. A further benefit is a faster loop: the response delay of the basic clamp (TVS avalanche build-up plus gate-node charging, typically 100–200 ns) allows a *dynamic* overshoot

$$\Delta V_{\mathrm{dyn}} \approx \left.\frac{\mathrm{d}v_{CE}}{\mathrm{d}t}\right|_{\mathrm{off}} \cdot t_{\mathrm{loop}} \tag{7}$$

above the static clamp level; with 2–4 kV/µs typical of this voltage class, 150 ns of loop delay costs 300–600 V and must be part of the voltage budget.

2.4 Relation to passive protections

Active clamping does not replace the decoupling snubber: the snubber handles the very fast local loop (module + busbar section, tens of nH) where the clamp loop is too slow, while the clamp handles the slow, high-energy part of the loop and abnormal events. Compared with an MOV, the clamp level is far better defined (±5% TVS tolerance versus the soft, tolerance- and temperature-affected V–I curve of a varistor), but the energy capability is bounded by the IGBT SOA, not by the clamp element; an MOV bank remains the right choice when kilojoules must be absorbed, as in the static-switch application of Section 3.5.

3 Design for the 6.5 kV / 1000 A Module

3.1 Ratings, assumptions and protection target

The design targets the Hitachi ABB HiPak 5SNA 1000G650300 [5]: $V_{CES} = 6.5\,\mathrm{kV}$, $I_C = 1000\,\mathrm{A}$, $I_{CM} = 2000\,\mathrm{A}$, gate limits ±20 V, driven at $+15/-10\,\mathrm{V}$. The following system assumptions are used in the worked example and should be replaced by the actual project values:

Quantity	Symbol	Value
Nominal DC link	V_{dc}	3600 V
Maximum DC link (abnormal)	$V_{dc,\max}$	4000 V
Maximum turn-off current	I_0	2000 A
External commutation loop inductance	L_σ	300 nH
Module internal inductance (order)	L_{int}	10 nH
Turn-off gate resistor (example)	$R_{g,\mathrm{off}}$	15 Ω
Negative gate supply	V_{EE}	−10 V
Miller plateau at I_0 (estimate)	$v_{GE,\mathrm{pl}}$	12 V
Ambient/board temperature range	—	$-25 \ldots +85\,^\circ\mathrm{C}$

The protection target is set at the module terminals:

$$v_{CE,\max} \le V_{\mathrm{lim}} = 6.0\,\mathrm{kV}, \tag{8}$$

reserving 500 V below V_{CES} for (i) the chip-level excess $L_{\mathrm{int}}\,|\mathrm{d}i/\mathrm{d}t|$ over the sensed terminal voltage, which with (3) is only $L_{\mathrm{int}}(v_{cl} - V_{dc})/L_\sigma \approx 60\,\mathrm{V}$ here but grows with snappier events, and (ii) the dynamic overshoot (7) of the clamp loop, which is the dominant term.

3.2 Threshold window

The stack breakdown voltage must satisfy a two-sided inequality over all tolerance and temperature corners:

$$N\,V_{BR,\min}\left(1+\alpha_T(T_{\min}-25\,^\circ\mathrm{C})\right) \;\geq\; V_{dc,\max}+\Delta V_{\mathrm{rep}}, \tag{9}$$

$$N\,V_{BR,\max}\left(1+\alpha_T(T_{\max}-25\,^\circ\mathrm{C})\right)+Nr_z i_{TVS}+v_{GE,\mathrm{pl}}+\Delta V_{\mathrm{dyn}} \;\leq\; V_{\mathrm{lim}}. \tag{10}$$

The lower bound (9) prevents static conduction at maximum DC link and, with the additional margin ΔV_{rep}, keeps the stack out of the *repetitive* turn-off overshoot so that it is only stressed by abnormal events. Avalanche diodes above ∼200 V have a positive temperature coefficient $\alpha_T \approx +1\cdot 10^{-3}\,/\mathrm{K}$, so the lower corner occurs cold and at the low tolerance limit, the upper corner hot and at the high limit.

3.3 TVS stack selection

A convenient building block is the bidirectional axial transil 1.5KE440CA [6]: $V_{BR} = 418\ldots462\,\mathrm{V}$ at 1 mA (±5%), stand-off 376 V with ≤ 1 µA leakage, $V_C = 602\,\mathrm{V}$ at $I_{PP} = 2.5\,\mathrm{A}$ (10/1000 µs), from which $r_z \approx (602-440)/2.5 \approx 65\,\Omega$. The bidirectional version is mandatory here: with the IGBT on, the gate (+15 V) sits above the collector (∼4 V) and a unidirectional stack would conduct forward from gate to collector; alternatively a series blocking diode can be added to a unidirectional stack.

Basic clamp, $N = 10$. The nominal threshold is 4.40 kV. The corners of (9)–(10) give:

	cold, low tol.	hot, high tol.
$V_{BR,s}$	$4180\times0.95 = 3.97\,\mathrm{kV}$	$4620\times1.06 = 4.90\,\mathrm{kV}$
$+\,Nr_z i_{TVS}$ ($i_{TVS}=1.5\,\mathrm{A}$)	—	$+0.98\,\mathrm{kV}$
$+\,v_{GE,\mathrm{pl}}$	—	$+0.01\,\mathrm{kV}$
result	3.97 kV vs. $V_{dc,\max}=4.0\,\mathrm{kV}$	5.89 kV vs. $V_{\mathrm{lim}}=6.0\,\mathrm{kV}$

with $i_{TVS} \approx (12+10)/15 \approx 1.5\,\mathrm{A}$ from (6). Both corners close, but with essentially no margin: the cold/low corner sits 30 V *below* the maximum DC link (acceptable only because near the knee the stack conducts sub-milliamp currents through $R_{g,\mathrm{off}}$ into the low-impedance driver output, but a standing dissipation and a slow gate lift must be checked), and the hot side leaves nothing for the dynamic overshoot (7). The basic scheme with this stack is therefore *marginal*: it requires either binned/measured TVS devices, a narrower temperature range, or a guaranteed $i_{TVS} \lesssim 1.5\,\mathrm{A}$.

Advanced clamp, $N = 11$. With an AAC driver input (Section 2.3) the sense current is $i_{TVS} \approx 0.3\,\mathrm{A}$ and one more device fits in the window:

	cold, low tol.	hot, high tol.
$V_{BR,s}$	$4598\times0.95 = 4.37\,\mathrm{kV}$	$5082\times1.06 = 5.39\,\mathrm{kV}$
$+\,Nr_z i_{TVS}$ ($i_{TVS}=0.3\,\mathrm{A}$)	—	$+0.21\,\mathrm{kV}$
$+\,v_{GE,\mathrm{pl}}$	—	$+0.01\,\mathrm{kV}$
result	$4.37\,\mathrm{kV} > 4.0\,\mathrm{kV}$ ✓	$5.61\,\mathrm{kV} \leq 6.0\,\mathrm{kV}$ ✓

Now 370 V of static margin remain at the bottom and ∼400 V at the top for the dynamic overshoot — consistent with 100–150 ns of loop delay at 3 kV/µs. This is the recommended configuration.

3.4 Energies and pulse stress (worked example)

With the nominal basic-clamp level $v_{cl} = 4400{+}975{+}12 \approx 5.4\,\mathrm{kV}$, $I_0 = 2000\,\mathrm{A}$ and $L_\sigma = 300\,\mathrm{nH}$:

$$\Delta t_{\mathrm{cl}} = \frac{300\,\mathrm{nH} \cdot 2000\,\mathrm{A}}{5.4\,\mathrm{kV} - 3.6\,\mathrm{kV}} \approx 0.34\,\mu\mathrm{s}, \tag{11}$$

$$E_{\mathrm{IGBT}} = \tfrac{1}{2} \cdot 300\,\mathrm{nH} \cdot (2000\,\mathrm{A})^2 \cdot \frac{5.4}{1.8} \approx 1.8\,\mathrm{J}, \tag{12}$$

$$E_{TVS} \approx 5.4\,\mathrm{kV} \cdot 1.5\,\mathrm{A} \cdot 0.34\,\mu\mathrm{s} \approx 2.8\,\mathrm{mJ} \quad (\approx 0.3\,\mathrm{mJ/device}), \tag{13}$$

$$P_{TVS,\mathrm{pk}} \approx (440 + 98)\,\mathrm{V} \cdot 1.5\,\mathrm{A} \approx 0.8\,\mathrm{kW/device\ for\ } 0.34\,\mu\mathrm{s}. \tag{14}$$

The single-event stress is trivial for a 1.5 kW-class transil (its short-pulse power capability is far above the 10/1000 µs rating). The 1.8 J absorbed by the IGBT add to the normal turn-off energy and are acceptable as an occasional event; if the clamp engaged at every cycle at this depth, even a few hundred hertz would add kilowatt-level losses and violate the SOA repetition assumptions — hence the repetitive-overshoot margin ΔV_{rep} in (9). Shallow grazing of the knee at the highest line voltage is tolerable and self-limiting (the TVS TC pushes the threshold up as the stack warms).

3.5 Coordination with the MOV clamp (static-switch case)

In the static-switch application [10] the turn-off of ∼3 kA against an RL load is intentionally performed against the counter-voltage of an EPCOS B80K680 MOV bank clamping at approximately 5.5 kV, which absorbs the multi-kilojoule load energy. An active clamp can only coexist with the MOV if its threshold lies *above* the MOV protection level at the peak current in all corners:

$$N\,V_{BR,\min}\,(1 + \alpha_T \Delta T_{\min}) \;\geq\; V_{\mathrm{MOV}}(\hat{I}) \approx 5.5\,\mathrm{kV} \quad \Rightarrow \quad N \geq \frac{5500}{418 \times 0.95} \approx 13.9. \tag{15}$$

With $N = 14$ the nominal threshold is already 6.16 kV and the clamp level exceeds V_{CES} even in the AAC variant: *no feasible window exists above a 5.5 kV MOV*. Conversely, a threshold below the MOV level is not admissible: during the millisecond-scale MOV conduction the clamp would hold the IGBT in linear conduction at ∼5 kV and steer the load current (and its kilojoules) into the silicon. The practical conclusions for the static switch are:

- the MOV bank remains the sole energy clamp; the fast local overshoot ahead of the MOV loop is handled by the per-module decoupling snubber, as already foreseen;
- an active clamp becomes feasible only if the MOV protection level is lowered (more parallel columns or a lower voltage class) to $\lesssim 4.6\,\mathrm{kV}$, opening an AAC window at $N = 12$–13 — at the cost of higher MOV standing voltage stress ratio;
- with several modules in parallel, per-module clamp networks with binned thresholds are required: the module with the lowest threshold takes the clamp energy first.

3.6 Implementation notes

- Sense at the *auxiliary* collector terminal so that the main-terminal inductance does not corrupt the feedback; keep the TVS loop short and away from the power loop.
- Add a small series resistor R_s (a few ohms to a few tens of ohms) to damp the interaction between the stack capacitance and the gate loop; verify it does not slow the clamp response unacceptably.
- Protect the gate with a local ±18 V G–E TVS pair: during clamp engagement the injected spike briefly overdrives the gate node.

- The engagement spike through the stack, dominated by C_{ies} recharge from V_{EE} to the plateau, is limited mainly by the stack dynamic resistance and R_s; check the instantaneous TVS current against the datasheet $I_{PP}(t_p)$ curve.
- The clamp interacts with the driver's protection functions: desaturation detection and soft turn-off must not fight the clamp (a soft-turn-off ramp during clamping is generally beneficial; a hard re-off pulse is not).
- In series- or parallel-connected assemblies the clamp also provides dynamic voltage balancing / current derating; matched thresholds across positions are then essential.

4 Simscape Model `tvs_clamp`

4.1 Scope and equations

The component `tvs_clamp` (Appendix A) models the complete stack — N series devices per string, M parallel strings — as a single electrical two-terminal, intended to be wired from the collector (auxiliary terminal) to the gate node of `igbt_dyn` in a double-pulse or application-level simulation. It follows the same conventions as the other components of the personal `.ssc` library.

The constitutive relation is a symmetric, purely algebraic piecewise-linear characteristic with a smoothed knee, in parallel with leakage and a lumped junction capacitance:

$$i = \frac{v}{R_{\mathrm{ls}}} + \frac{\mathrm{sp}(v - V_{\mathrm{brs}}) - \mathrm{sp}(-v - V_{\mathrm{brs}})}{r_{\mathrm{zs}}} + C_s \frac{\mathrm{d}v}{\mathrm{d}t}, \tag{16}$$

$$\mathrm{sp}(x) = \tfrac{1}{2}\left(x + \sqrt{x^2 + \Delta V_s^2}\right), \tag{17}$$

with the stack equivalents

$$V_{\mathrm{brs}} = N V_{BR}\big(1 + \alpha_T (T_{\mathrm{dev}} - T_{\mathrm{ref}})\big), \quad r_{\mathrm{zs}} = \frac{N r_z}{M}, \quad R_{\mathrm{ls}} = \frac{N R_{\mathrm{leak}}}{M}, \quad C_s = \frac{M C_j}{N}, \quad \Delta V_s = N\,\Delta V. \tag{18}$$

The function $\mathrm{sp}(x)$ is a softplus-like smooth positive part: it equals $\max(0, x)$ away from the knee and rounds it over a width of order ΔV_s. Being algebraic (no exponentials), it cannot overflow and behaves well with stiff solvers — the same reasoning used in the depletion clamps of `diode_rr`. The two mirrored terms in (16) make the characteristic odd, i.e. the stack is bidirectional, matching the CA-suffix devices; for a unidirectional stack with series blocking diode, the negative branch can be removed and a forward-diode branch added.

4.2 Parameters and defaults

Defaults describe the 1.5KE440CA:

Parameter	Symbol	Default	Source / note
`N`	N	10	series devices per string
`M`	M	1	parallel strings
`Vbr`	V_{BR}	440 V	±5% @ 1 mA [6]
`rz`	r_z	65 Ω	$(V_C - V_{BR})/I_{PP} = (602 - 440)/2.5$
`Rleak`	R_{leak}	400 MΩ	376 V / ≤ 1 µA
`Cj`	C_j	100 pF	order of magnitude; take from datasheet
`dV`	ΔV	8 V	knee smoothing per device
`alphaT`	α_T	$1 \cdot 10^{-3}$ /K	avalanche TC, $V_{BR} > 200$ V
`Tdev`, `Tref`	$T_{\mathrm{dev}}, T_{\mathrm{ref}}$	25 °C	fixed device temperature

Tolerance corners are explored by scaling `Vbr` to 418/462 V and sweeping `Tdev`; there is deliberately no Monte Carlo machinery inside the component.

4.3 Limitations

(i) No self-heating: V_{BR} shifts during a long clamp pulse are not reproduced (conservative for the upper corner if T_{dev} is set hot). (ii) r_z is constant, while real transils are sublinear at high pulse current; near and above I_{PP} the model overestimates the clamp voltage — acceptable, since the design intent is precisely to keep i_{TVS} small. (iii) The avalanche build-up delay (ns range) is neglected; the dominant loop delay comes from the gate node and must be represented by the driver/gate model, not by the TVS.

4.4 Suggested validation

Reuse the double-pulse bench built around `igbt_dyn` with the 5SNA 1000G650300 parameter set: connect `tvs_clamp` from the collector node to the gate node behind $R_{g,\mathrm{off}}$, drive with the ideal $+15/-10\,\mathrm{V}$ source, and sweep I_0 (500–2000 A) and L_σ (100–600 nH). Check: (a) the v_{CE} plateau against (2), (b) Δt_{cl} and the current slope against (3), (c) i_{TVS} against (6) including the engagement spike, (d) energies via integrated $v{\cdot}i$ against (4)–(5). If the solver struggles at the knee, increase `dV` (the clamp level error is of order $\Delta V_s/2$ at most) before touching tolerances, as done for the tail-charge equations of `igbt_dyn`.

5 Conclusion

Active clamping turns the 6.5 kV IGBT itself into its own overvoltage absorber at the cost of a signal-level TVS network. For the 5SNA 1000G650300 on a 3.6 kV link, a stack of ten 1.5KE440CA devices closes the design window only marginally in the basic scheme, while eleven devices with an advanced (driver-assisted) clamp input leave healthy static and dynamic margins; in the MOV-clamped static switch no admissible window exists above the 5.5 kV varistor level, and the MOV must remain the sole energy clamp. The `tvs_clamp` Simscape component captures the stack with a numerically robust smoothed breakdown characteristic and slots directly into the existing `igbt_dyn` double-pulse bench for time-domain verification of clamp level, timing and energies.

References

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[3] ABB Switzerland Ltd, Semiconductors, "Applying IGBTs," Application Note 5SYA 2053.

[4] CT-Concept Technologie AG / Power Integrations, application literature on the Advanced Active Clamping function of SCALE gate-driver cores for high-voltage IGBTs.

[5] Hitachi ABB Power Grids, "5SNA 1000G650300 HiPak IGBT Module," Data Sheet 5SYA 1465.

[6] STMicroelectronics / Vishay, "1.5KE series — 1500 W Transil / TVS diodes," data sheet.

[7] D. Bagnara, *Snubber Design Guidelines for a High-Current IGBT Half-Bridge Leg*, internal note.

[8] D. Bagnara, *Snubbers for Paralleled 5SNA 1000G650300 IGBT Modules*, internal note.

[9] D. Bagnara, *The EPCOS SIOV-B80K680 Block Varistor: Characteristics and Application Sizing*, internal note.

[10] D. Bagnara, *High-Current Static Switch — Preliminary Proposal*, internal note.

A Source of `tvs_clamp.ssc`

```
component tvs_clamp
% TVS Clamp Stack
% Bidirectional TVS (transil) stack for IGBT active clamping studies:
% N series devices per string, M parallel strings, modelled as a single
% two-terminal element. Symmetric avalanche breakdown with a smooth,
% purely algebraic knee (no exponentials), constant dynamic resistance,
% sub-breakdown leakage and a lumped junction capacitance. The breakdown
% voltage has a linear temperature coefficient; the device temperature is
% a fixed parameter (no self-heating).
%
% Default parameters: 1.5KE440CA (V_BR = 440 V +/-5% @ 1 mA,
% V_C = 602 V @ I_PP = 2.5 A -> r_z ~ 65 Ohm, V_RM = 376 V / I_R <= 1 uA
% -> R_leak ~ 400 MOhm). Intended use: collector-to-gate clamp network
% together with igbt_dyn (e.g. 5SNA 1000G650300 parameter set).

  nodes
    p = foundation.electrical.electrical; % +:left
    n = foundation.electrical.electrical; % -:right
  end

  parameters
    N      = {10,      '1'   }; % Series devices per string, N
    M      = {1,       '1'   }; % Parallel strings, M
    Vbr    = {440,     'V'   }; % Breakdown voltage per device @ Tref, V_BR
    rz     = {65,      'Ohm' }; % Dynamic resistance per device, r_z
    Rleak  = {400e6,   'Ohm' }; % Sub-breakdown leakage resistance per device
    Cj     = {100e-12,'F'   }; % Junction capacitance per device, C_j
    dV     = {8,       'V'   }; % Knee smoothing voltage per device
    alphaT= {1e-3,    '1/K' }; % Breakdown temperature coefficient, alpha_T
    Tdev   = {298.15, 'K'   }; % Device temperature
    Tref   = {298.15, 'K'   }; % Reference temperature for V_BR
  end

  parameters (Access = private)
    Vbrs = N*Vbr;       % nominal stack breakdown (before TC correction)
    rzs  = N*rz/M;      % stack dynamic resistance
    Rls  = N*Rleak/M;   % stack leakage resistance
    Cs   = M*Cj/N;      % stack capacitance
    dVs  = N*dV;        % stack knee smoothing voltage
  end

  variables
    v = {0, 'V'}; % Voltage
    i = {0, 'A'}; % Current
  end

  branches
    i : p.i -> n.i;
  end

  intermediates
    % Temperature-corrected stack breakdown voltage
    VbrT = Vbrs*(1 + alphaT*(Tdev - Tref));
    % Smooth positive-part (softplus-like, algebraic: max(0,x) with
    % rounding over ~dVs around the knee, safe for stiff solvers)
```

```
    xp  = v - VbrT;
    xn  = -v - VbrT;
    spp = 0.5*(xp + sqrt(xp^2 + dVs^2));
    spn = 0.5*(xn + sqrt(xn^2 + dVs^2));
  end

  equations
    v == p.v - n.v;
    i == v/Rls + (spp - spn)/rzs + Cs*der(v);
  end

end
```